

2020 AMC 10A Problem 1 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 1. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 1

Problem:

What value of x satisfies

$$x - \frac{3}{4} = \frac{5}{12} - \frac{1}{3}?$$

Answer Choices:

A. $-\frac{2}{3}$

B. $\frac{7}{36}$

C. $\frac{7}{12}$

D. $\frac{2}{3}$

E. $\frac{5}{6}$

Solution:

Adding $\frac{3}{4}$ to both sides:

$$\begin{aligned}x &= \frac{5}{12} - \frac{1}{3} + \frac{3}{4} \\&= \frac{5}{12} - \frac{4}{12} + \frac{9}{12} \\&= (E) \boxed{\frac{5}{6}}\end{aligned}$$

OR

Multiplying 12 on both sides gets us:

$$\begin{aligned}12x - 9 &= 1 \\ \implies 12x &= 10\end{aligned}$$

Therefore $x = (E) \boxed{\frac{5}{6}}$.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

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